

# Empirical Evaluation & Benchmark Results

**Evaluation Mode:** Real Empirical Run (Zero Fabrication Guarantee)  
**Dataset:** data/processed/processed\_flows.csv (Holdout Split: 20% Stratified)  
**Evaluated Date:** 2026-08-18 13:35:33

## 1. Multi-Threshold Performance (Supervised Flow Classifier)

The classifier was evaluated across multiple probability cutoffs. In production NDR, we employ a **Precision-First posture (threshold = 0.85)** to prevent alert fatigue while delegating evasive/novel patterns to the Suricata signature and Autoencoder layers.

| Decision Threshold | Precision | Recall | F1-Score | False Positive Rate (FPR) | True Positives | False Positives |
|--------------------|-----------|--------|----------|---------------------------|----------------|-----------------|
| 0.50               | 0.9998    | 0.9996 | 0.9997   | 0.0004                    | 12234          | 3               |
| 0.70               | 0.9998    | 0.9996 | 0.9997   | 0.0004                    | 12234          | 3               |
| 0.85               | 0.9998    | 0.9995 | 0.9996   | 0.0004                    | 12233          | 3               |
| 0.95               | 0.9999    | 0.9990 | 0.9995   | 0.0001                    | 12227          | 1               |

## 2. Per-Category Classification Breakdown (Threshold = 0.85)

| Attack Category | Precision | Recall | F1-Score | Support |
|-----------------|-----------|--------|----------|---------|
| BENIGN          | 0.9994    | 0.9996 | 0.9995   | 7763    |
| DDOS_FLOOD      | 0.9998    | 0.9996 | 0.9997   | 12239   |
| macro avg       | 0.9996    | 0.9996 | 0.9996   | 20002   |
| weighted avg    | 0.9996    | 0.9996 | 0.9996   | 20002   |

## 3. Unsupervised Autoencoder Anomaly Layer

Trained strictly on benign traffic flows with reconstruction error threshold calibrated at **1865.197388**.

| Metric            | Measured Value |
|-------------------|----------------|
| Anomaly Precision | 0.0000         |
| Anomaly Recall    | 0.0000         |
| Anomaly F1-Score  | 0.0000         |
| Anomaly FPR       | 0.0116         |
| True Negatives    | 7673           |
| False Positives   | 90             |